

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026

Biology — Class IX | Paper: SSC-I

Syllabus: Ch 1 — The Science of Biology | Ch 2 — Biodiversity | Ch 3 — The Cell | Ch 4 — Cell Cycle | Ch 5 — Tissues, Organs & Organ Systems | Ch 6 — Molecular Biology | Ch 7 — Metabolism (Sec 7.1-7.6)

Total Marks: 65

Time: 2h 30m

Version: 1

General Instructions:

1. Write your Roll Number in the space provided on the answer sheet. Do NOT write your name.
2. Attempt ALL questions from Section A. In Sections B and C, attempt the required number of questions, choosing either the main question OR its alternative (OR).
3. All parts of a question must be answered in sequence and at one place.
4. Extra attempt of any question or any part thereof will NOT be considered.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Question	A	B	C	D	Answer
1	Which step of the biological method does the textbook describe as the MOST important step?	☐ Observation	☐ Formulation of hypothesis	☐ Experimentation	☐ Summarization of results	
2	A statement that no experiment could ever possibly prove wrong fails which required characteristic of a hypothesis?	☐ It should be a general statement	☐ It should be a tentative idea	☐ It should agree with available observations	☐ It should be testable and potentially falsifiable	
3	According to the textbook, prions are composed of:	☐ Circular RNA only	☐ Proteins only	☐ DNA only	☐ A protein coat around a DNA core	
4	In the five-kingdom classification system, unicellular prokaryotic organisms are placed in kingdom:	☐ Protista	☐ Fungi	☐ Monera	☐ Plantae	
5	Which of the following is NOT present in a prokaryotic cell?	☐ Cell membrane	☐ Ribosomes	☐ Cytoplasm	☐ Membrane-bound nucleus	
6	In which phase of meiosis do the bivalents line up at the equatorial plate of the spindle?	☐ Prophase I	☐ Metaphase I	☐ Anaphase I	☐ Metaphase II	
7	A tumour made of normal cells that stays in one place and does not spread to other parts of the body is called:	☐ Malignant	☐ Metastatic	☐ Benign	☐ Cancerous	
8	In the levels of organization, two or more types of tissues that work together to complete a specific task form a(n):	☐ Organelle	☐ Organ	☐ Organ system	☐ Organism	
9	Which of the three main plant tissue types covers and protects the plant?	☐ Ground tissue	☐ Vascular tissue	☐ Dermal tissue	☐ Meristematic tissue	
10	The disaccharide that is present in milk is:	☐ Sucrose	☐ Maltose	☐ Lactose	☐ Glycogen	
11	The process in which the base sequence of a DNA strand is copied into a molecule of mRNA is called:	☐ Replication	☐ Transcription	☐ Translation	☐ Condensation	
12	A detachable inorganic ion such as Zn^{++} or a chloride ion, which an enzyme needs in order to work properly, is called a(n):	☐ Activator	☐ Coenzyme	☐ Prosthetic group	☐ Inhibitor	

SECTION B — Short Questions (11 × 3 = 33 Marks)

Attempt 11 questions. Choose either the main question OR its alternative.

Part	Question	Marks	OR	Alternative Question	Marks
2(i)	State the FOUR characteristics that a hypothesis should have, according to your textbook.	3	OR	Observations are of two types. Distinguish between qualitative and quantitative observations, giving ONE example of each, and state which type is more accurate and why.	3
2(ii)	Viruses show the characteristics of both living and non-living things. State TWO living characteristics and TWO non-living characteristics of viruses, and explain why viruses are not placed in any domain or kingdom.	3	OR	What are prions and viroids? State what each of them is composed of, and name ONE characteristic that prions, viroids and viruses all share.	3
	Name the FIVE kingdoms of the five-kingdom			Define saprotroph, autotroph and heterotroph.	

2(iii)	classification system. State which of them contains prokaryotes, and which contains eukaryotes that have a cell wall but cannot perform photosynthesis.	3	OR	For each mode of nutrition, name ONE kingdom of domain Eukarya in which it is found.	3
2(iv)	Muscle cells are specialized for contraction. Name the THREE types of muscle cell shown in Fig 3.21 of your textbook, state ONE place in the body where each type is found, and name the protein that fills muscle cells so that they can contract.	3	OR	Red blood cells are an example of cell specialization. State TWO structural features of a red blood cell and explain how each feature helps it to carry out its function of transporting oxygen.	3
2(v)	Why is prophase I described as the longest phase of meiosis? Describe THREE events that take place during prophase I.	3	OR	Describe what happens during metaphase I and during anaphase I of meiosis. If the parent cell has a diploid number of $2n = 12$, state how many chromosomes each pole receives at the end of anaphase I.	3
2(vi)	Study the figure printed below this row. It shows the cell cycle of an animal cell (Fig 4.3, adapted from your textbook) with the names of the five stages removed. Identify the stages labelled A, B, C, D and E.	3	OR	What is a benign tumour, and how does it differ from a malignant tumour? Explain what is meant by metastasis, and state what goes wrong in a cell for a tumour to form in the first place.	3

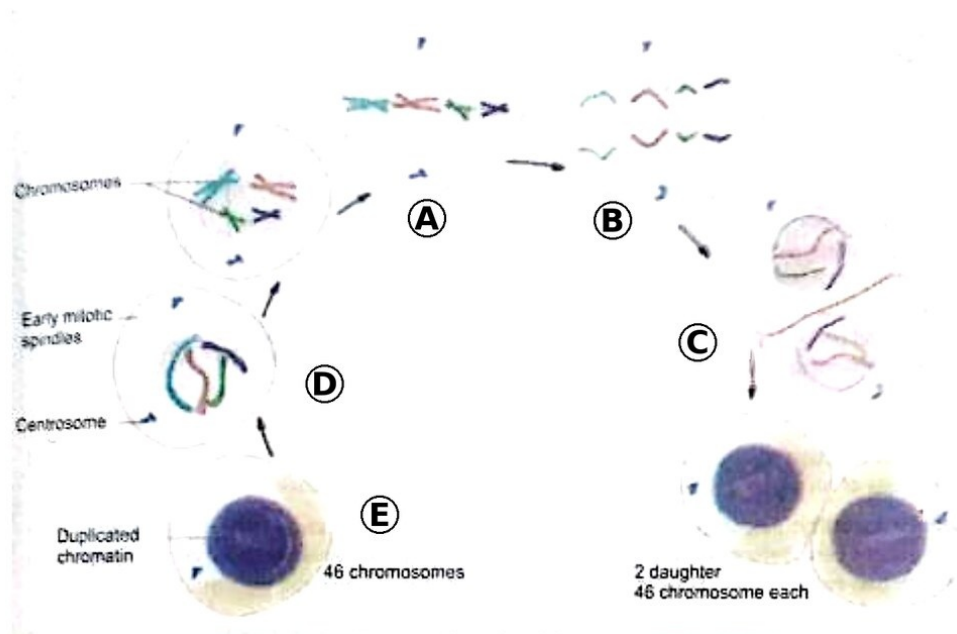


Fig 4.3: Mitosis, with the stage names removed — for use with Q 2(vi)

A = _____ B = _____ C = _____ D = _____ E = _____

2(vii)	List the FIVE levels of biological organization from the cellular level up to the organism level, and give ONE example from your textbook at each level.	3	OR	Name the FOUR types of animal tissue described in your textbook and state ONE function of each.	3
2(viii)	Plant tissues differentiate into three main types. Name them and state the function of each.	3	OR	With reference to Fig 5.2 of your textbook, explain why the stomach is described as an organ rather than a tissue. Name the tissue types found in its layered walls.	3
2(ix)	Name the TWO monosaccharide units that join together by condensation to form each of the following disaccharides: (a) sucrose, (b) maltose, (c) lactose. State what else is released each time the bond forms.	3	OR	Cellulose and starch are both polysaccharides made from glucose. State ONE function of cellulose in a plant, ONE use of cellulose to humans, and explain why cellulose is still important in the human diet even though we cannot digest it.	3
2(x)	Describe the process of transcription. State where in the cell it takes place, what acts as the template, and name the molecule that is produced.	3	OR	Describe the process of translation. State where in the cell it takes place, the role of tRNA, and what is produced at the end.	3
2(xi)	Many enzymes cannot work on their own and need a non-protein component called a co-factor. Name the THREE types of co-factor described in your textbook and give ONE example of each.	3	OR	Explain why only a very small quantity of an enzyme is needed to catalyze a large amount of substrate. In your answer, state what happens to the enzyme molecule during the reaction and what happens to it afterwards.	3

SECTION C — Long Questions (4 × 5 = 20 Marks)

Attempt all 4 questions. Choose either the main question OR its alternative.

Part	Question	Marks	OR	Alternative Question	Marks
3(i)	A student notices that the plants growing along one side of the school garden are much smaller and paler than the plants of the same kind growing along the other side. Using the SIX steps of the biological method in the correct order, show how the student would investigate this problem. Write at least one sentence for each step, making clear what the hypothesis would be, what deduction follows from it, and how the experiment would test that deduction.	5	OR	(a) State the FOUR characteristics of a good hypothesis. (b) Explain what is meant by a deduction, and show with ONE worked example how "if ... then" logic is used to draw a deduction from a hypothesis. (c) Explain why experimentation is called the most important step of the biological method.	5
3(ii)	Describe the FOUR stages of meiosis II (prophase II, metaphase II, anaphase II and telophase II), stating the main event that takes place in each. Then state how many daughter cells exist at the end of meiosis II, whether they are haploid or diploid, and state whether any DNA replication takes place during interphase II.	5	OR	Draw a well-labelled diagram of a homologous pair of chromosomes during prophase I, showing crossing over between non-sister chromatids (as in Fig 4.9 of your textbook). Label the homologous chromosomes, the non-sister chromatids and a chiasma. Then explain what a chiasma is, and state why crossing over matters to a species.	5
3(iii)	Describe the flow of genetic information from DNA to protein. Name the TWO stages involved, describe what happens in each, state where in the cell each one takes place, and name the type of RNA that carries the message and the type that brings the amino acids.	5	OR	Explain how hereditary information is passed from one generation to the next. In your answer: (a) state the relationship between a chromosome, DNA and a gene; (b) explain how gametes act as the vehicle for transferring hereditary information; and (c) explain why offspring resemble their parents in many characters and yet are not identical to them.	5
3(iv)	A pharmaceutical company is designing a medicine that must switch off an enzyme belonging to a disease-causing bacterium. Explain the TWO types of enzyme inhibition described in your textbook. For each type state where the inhibitor binds, what happens to the active site, and whether the effect can be overcome by increasing the substrate concentration. Finally, state which of the two types would be the better basis for such a medicine, and give a reason for your choice.	5	OR	(a) Explain what is meant by saying that an enzyme is specific, using amylase as your example. (b) State the optimum pH of pepsin and of trypsin, and explain why the two values are so different. (c) Explain what happens to an enzyme molecule at a temperature far above its optimum and why this effect is permanent, and contrast this with what happens at a temperature far below its optimum.	5

Invigilator's Signature: _____

Candidate's Signature: _____

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